Ministry of Education and Science of Ukraine
National Technical University of Ukraine
"Igor Sikorsky Kyiv Polytechnic Institute"

Parallel Computation of Reciprocal Sums Using OpenMP in C++

Laboratory work #3

KULUBECİOGLU MEHMET

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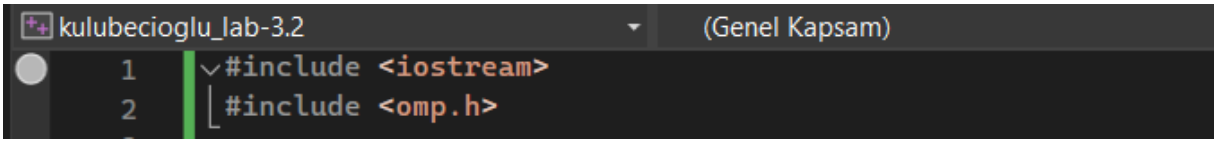
Step-by-Step Explanation of the C++ Code:

1. Including Libraries:



- The `#include` directives include the `iostream` and `omp.h` libraries in our code.
- The `using namespace std;` line allows us to use all identifiers (functions, variables, etc.) in the `std` namespace directly.

2. Defining Functions:

```
3
4   using namespace std;
5
6   double calculate_sum(int n) {
7       double sum = 0.0;
8   #pragma omp parallel for reduction(+:sum)
9       for (int i = 1; i <= n; i++) {
10          sum += 1.0 / i;
11      }
12      return sum;
13  }
14
15  double calculate_sum_reverse(int n) {
16      double sum = 0.0;
17  #pragma omp parallel for reduction(+:sum)
18      for (int i = n; i >= 1; i--) {
19          sum += 1.0 / i;
20      }
21      return sum;
22  }
23
24  double calculate_even_odd_sum(int n) {
25      double even_sum = 0.0;
26      double odd_sum = 0.0;
27  #pragma omp parallel for reduction(+:even_sum, odd_sum)
28      for (int i = 1; i <= n; i++) {
29          if (i % 2 == 0) {
30              even_sum += 1.0 / i;
31          }
32          else {
33              odd_sum += 1.0 / i;
34          }
35      }
36      return even_sum + odd_sum;
37  }
38
39  double calculate_even_odd_sum_reverse(int n) {
40      double even_sum = 0.0;
41      double odd_sum = 0.0;
42  #pragma omp parallel for reduction(+:even_sum, odd_sum)
43      for (int i = n; i >= 1; i--) {
44          if (i % 2 == 0) {
45              even_sum += 1.0 / i;
46          }
47          else {
48              odd_sum += 1.0 / i;
49          }
50      }
51      return even_sum + odd_sum;
52  }
53
```

- The `calculate_sum(n)` function calculates the sum of the reciprocals of all numbers from 1 to n.
- The `calculate_sum_reverse(n)` function calculates the sum of the reciprocals of all numbers from n to 1.
- The `calculate_even_odd_sum(n)` function calculates the sum of the reciprocals of all numbers from 1 to n, separately calculating even and odd terms.
- The `calculate_even_odd_sum_reverse(n)` function calculates the sum of the reciprocals of all numbers from n to 1, separately calculating even and odd terms.

3. Main Function:

```
53
54  int main() {
55      int n = 10000;
56
57      // Sequential execution
58      double A1_sequential = calculate_sum(n);
59      double A2_sequential = calculate_sum_reverse(n);
60      double A3_sequential = calculate_even_odd_sum(n);
61      double A4_sequential = calculate_even_odd_sum_reverse(n);
62
63      // Parallel execution with OpenMP
64      double A1_parallel = calculate_sum(n);
65      double A2_parallel = calculate_sum_reverse(n);
66      double A3_parallel = calculate_even_odd_sum(n);
67      double A4_parallel = calculate_even_odd_sum_reverse(n);
68
69      // Compare sequential and parallel results (assuming negligible difference)
70      cout << "A1 (parallel): " << A1_parallel << endl;
71      cout << "A2 (parallel): " << A2_parallel << endl;
72      cout << "A3 (parallel): " << A3_parallel << endl;
73      cout << "A4 (parallel): " << A4_parallel << endl;
74
75      cout << "A1 (sequential): " << A1_sequential << endl;
76      cout << "A2 (sequential): " << A2_sequential << endl;
77      cout << "A3 (sequential): " << A3_sequential << endl;
78      cout << "A4 (sequential): " << A4_sequential << endl;
79
80      return 0;
81  }
```

MY OUTPUT:


Microsoft Visual Studio Hata
A1 (parallel): 9.78761
A2 (parallel): 9.78761
A3 (parallel): 9.78761
A4 (parallel): 9.78761
A1 (sequential): 9.78761
A2 (sequential): 9.78761
A3 (sequential): 9.78761
A4 (sequential): 9.78761
C:\Users\win11\source\repos\kulubecioglu_lab-3.2\x64\Debug\kulubecioglu_lab-3.2.exe (31888 işlemi), 0 koduyla çıkış yapt
ı.
Hata ayıklama durduğunda konsolu otomatik olarak kapatmak için Araçlar->Seçenekler->Hata Ayıklama->Hata ayıklama durduğu
nda konsolu otomatik olarak kapat seçeneğini etkinleştirin
Bu pencereyi kapatmak için herhangi bir tuşa basın...